MOSbius Workshop Manual

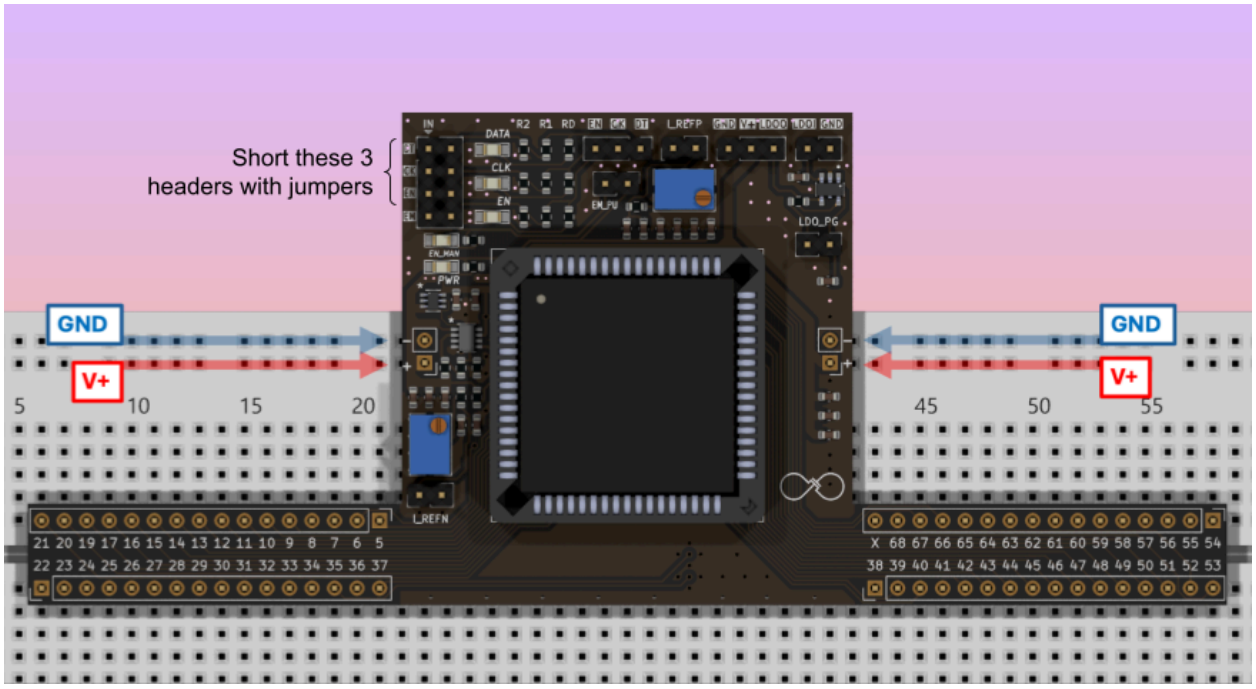
Overview
This workshop provides a hands-on introduction to designing, programming, and testing circuits using the MOSbius platform. MOSbius is an educational platform designed to give students practical experience with transistor-level MOS circuit design. The MOSbius chip is fabricated in a 65 nm CMOS technology and includes configurable nMOS and pMOS transistors, along with common analog circuit building blocks that can be connected to create and test different circuits via an onboard, programmable switch matrix. Participants will set up the MOSbius hardware paired with a Raspberry Pi Pico, use the MOSbius circuit simulator to design a real circuit, and then program and verify the design on the MOSbius chip.

By the end of the workshop, participants will gain experience with the basic MOSbius design workflow, hardware setup, and simulation.

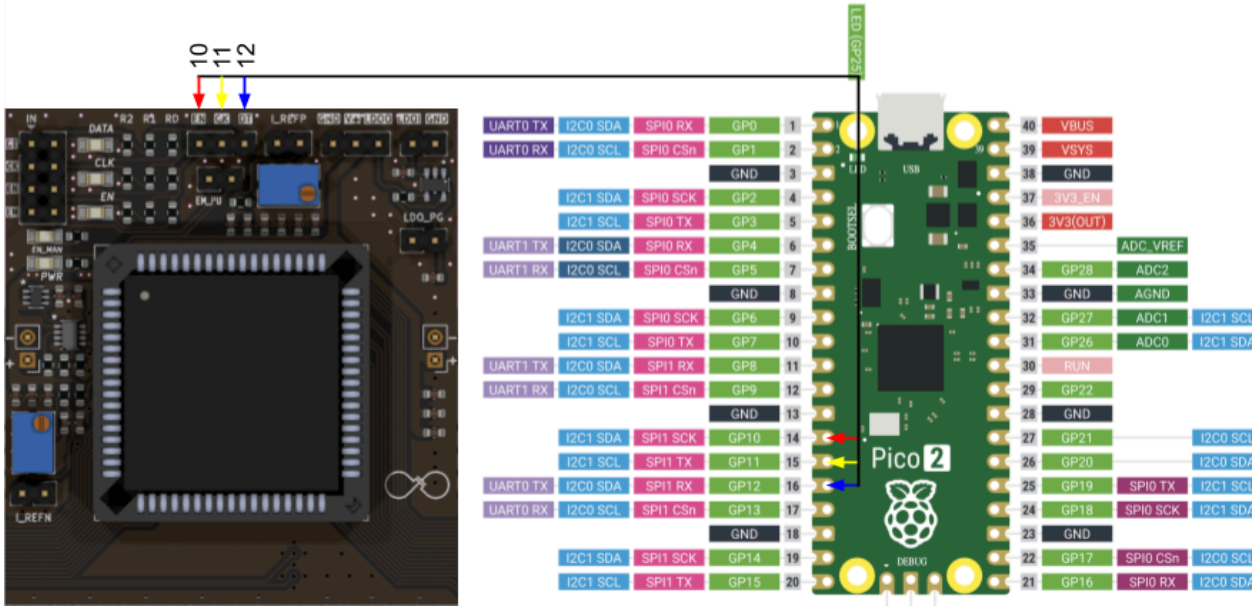
Prerequisites
- MOSbius
- Breadboard
- Raspberry Pi Pico 2
- Jumper wires
- Jumper shunts
- Resistors
- Capacitors
- USB-C adapter
- Laptop

Hardware Setup
Programming the MOSbius chip involves using the Raspberry Pi Pico microcontroller. We will be using the Pico to generate the digital control signals required to program and configure the MOSbius chip. Additionally, the Pico will provide power to the MOSbius board. When the Pico is connected through USB, it receives a 5V supply, which is regulated down to 3.3V by the Pico's onboard regulator. This 3.3 V output will be used to power the MOSbius chip.

To configure the platform for programming with the Pico, short each of the CT, CK, and EN headers with a jumper, as indicated in the following diagram:

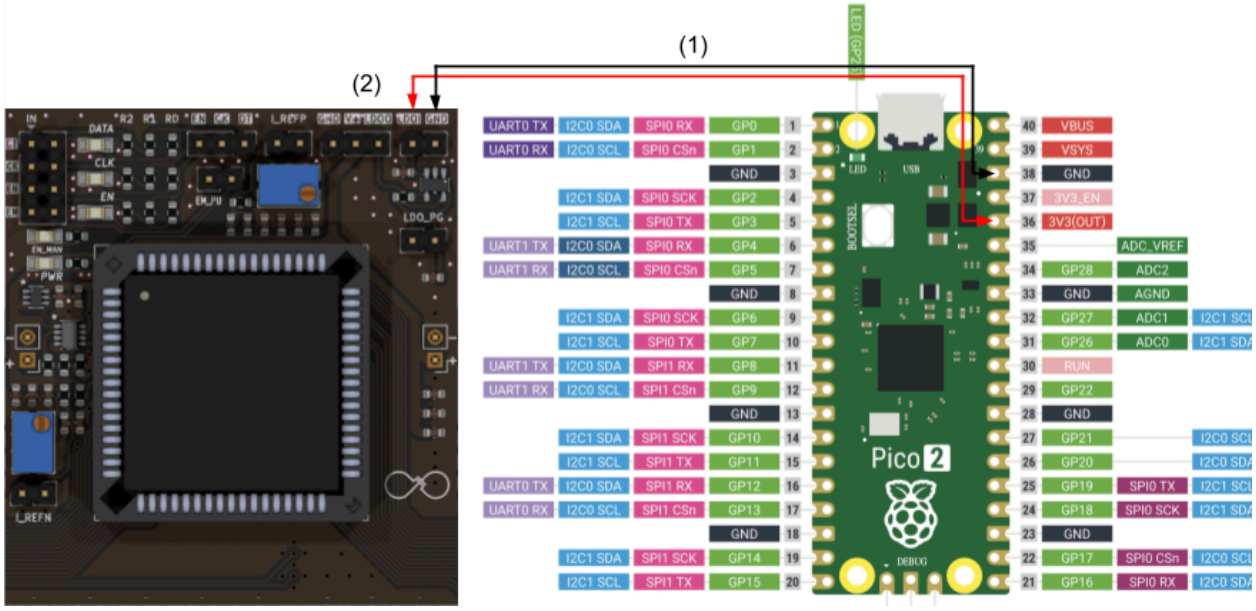


Connect GPIO pins 10, 11, and 12 on the Pico to the EN, CK, and DT pins on the MOSbius, respectively. The following diagram shows the connection points:



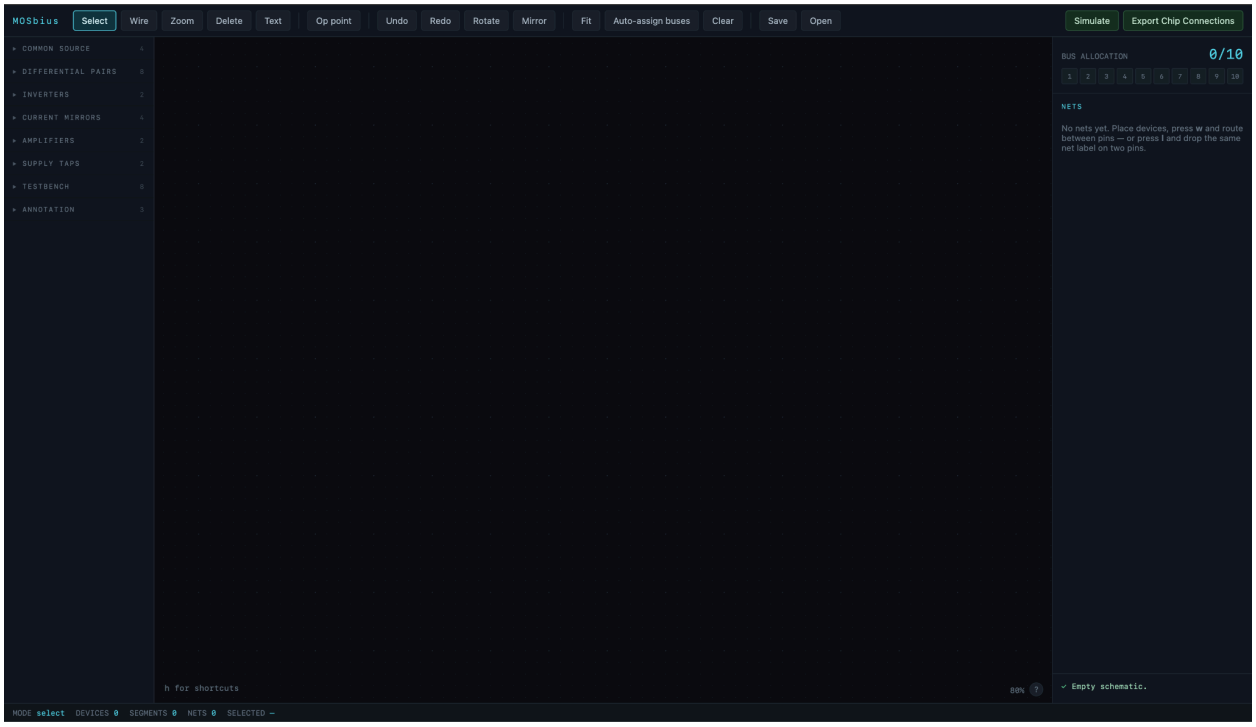
To power the MOSbius chip, we will be using the onboard LDO located on the MOSbius PCB to regulate the 3.3V output of the Pico to 2.5V. (1) Connect pin 36 (3V3/OUT) of the Pico to LDOI (LDO input) on the PCB. Additionally, tie any of the Pico's ground pins (3, 8, 13, 18, 23, 28, 33, or 38) to GND on the PCB. (2) Plug in the Pico to your laptop using the USB-C adapter. Verify using a voltmeter that LDOO (LDO output) is ~2.5V relative to GND. Once you have verified

LDOO, place a jumper across V+ and LDOO to power the MOSbius chip. Verify that the red PWR LED lights up on the MOSbius PCB.



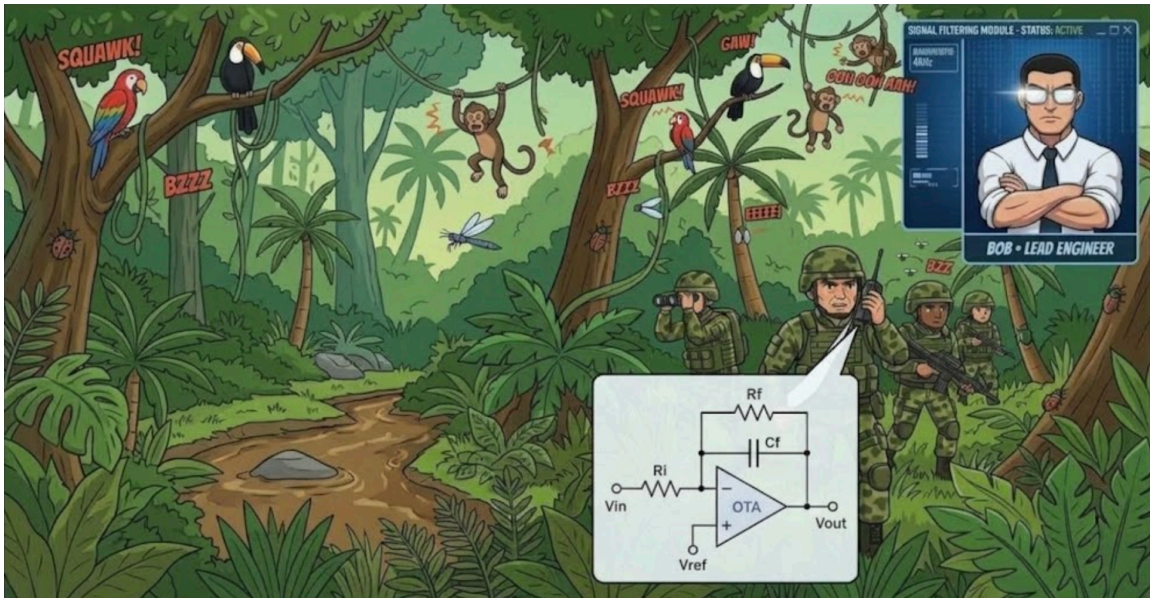
Software Setup

A custom local HTML webpage was designed to simplify the design, simulation, and programming process. The HTML can be downloaded from this Github repository (either via a clone or direct download). Launch the page by double clicking the downloaded HTML file. It should open automatically in your default browser.



The panel on the left contains the components, which you can click to place anywhere in the schematic window in the center of the window. Once you've finished laying out your schematic, you can click "Simulate" to run a simulation. To program the chip, first assign the switch matrix buses by clicking "Auto-assign buses", then press "Export Chip Connections", and finally click "Program Raspberry Pi Pico".

LAB 1: Active Lowpass Filter



Scenario:
Your team is building walkie-talkies for a special-forces unit heading into the jungle on a scout mission. The jungle is loud: birds, insects and monkeys fill the air with high-pitched noise that drowns out radio voice traffic. Your lead engineer, Bob, needs a filter that keeps human speech, strips out the high-frequency jungle noise, and boosts the voice signal before it's transmitted. Human speech carries almost all its understandable content below about 4 kHz.

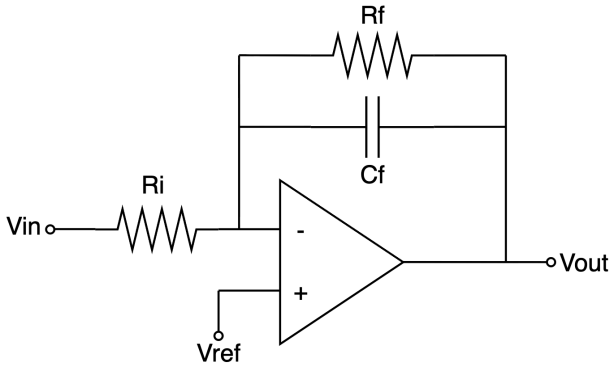
Your Task:
Design an active low-pass filter (assuming infinite open-loop gain) using the MOSbius on-chip OTAn that meets these specifications:

Specification Table	
Gain	20 dB
Bandwidth	4 kHz

Design Constraints:
You must use the following:

NMOS Current Mirror Array Bias	100 µA
Amplifier	Pre-built OTAn
Feedback Capacitor	50 pF
Supply Voltage	2.5 V
Input Common-Mode Voltage	1.6 V

Design Guide:



Start by assuming an ideal op-amp with infinite open-loop gain. An ideal op-amp in a negative feedback configuration will output whatever voltage it takes to make its two inputs equal. So the negative input terminal sits exactly at V_{ref}. This is called the virtual ground.

For this exercise, since the feedback capacitor value is constrained, the bandwidth sets the feedback resistor value while the gain sets the input resistor value.

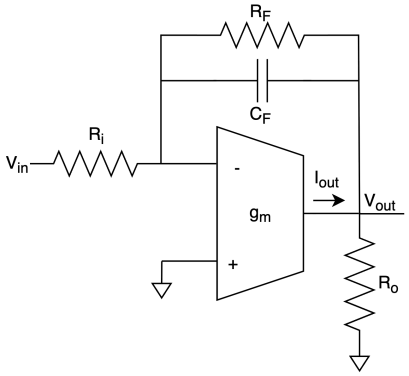
$$f_c = \frac{1}{2\pi R_f C_f} \qquad\qquad G = -\frac{R_f}{R_i}$$

Using the equations above, choose R_f and R_i such that the active lowpass filter meets the required specifications (convert the gain to V/V).

Simulation Results:

Build the circuit using the OTA nMOS symbol and tie its pin 19 to the pin 19 on the nMOS mirror array. Run an AC simulation and measure the in-band gain and the frequency where the gain drops by 3 dB. Record both values on the scratch paper provided. Do they match your hand calculations?

So far, we have treated this OTA like an ideal op-amp with infinite gain. However, a real OTA works differently. It converts the voltage difference between its inputs into an output current, $I_{out}=g_m(V^+ - V^-)$ which flows into its own output resistance R_o (and the feedback network). Here we also drop the assumption of infinite gain, meaning V^- no longer sits exactly at V^+. The OTA schematic is shown below.



Writing KCL at the V^- node and at the output node—solving it gives:

$$\frac{V_o}{V_{in}} = \frac{R_o - g_m R_o Z_F}{(1 + g_m R_o) R_{in} + R_o + Z_F}$$

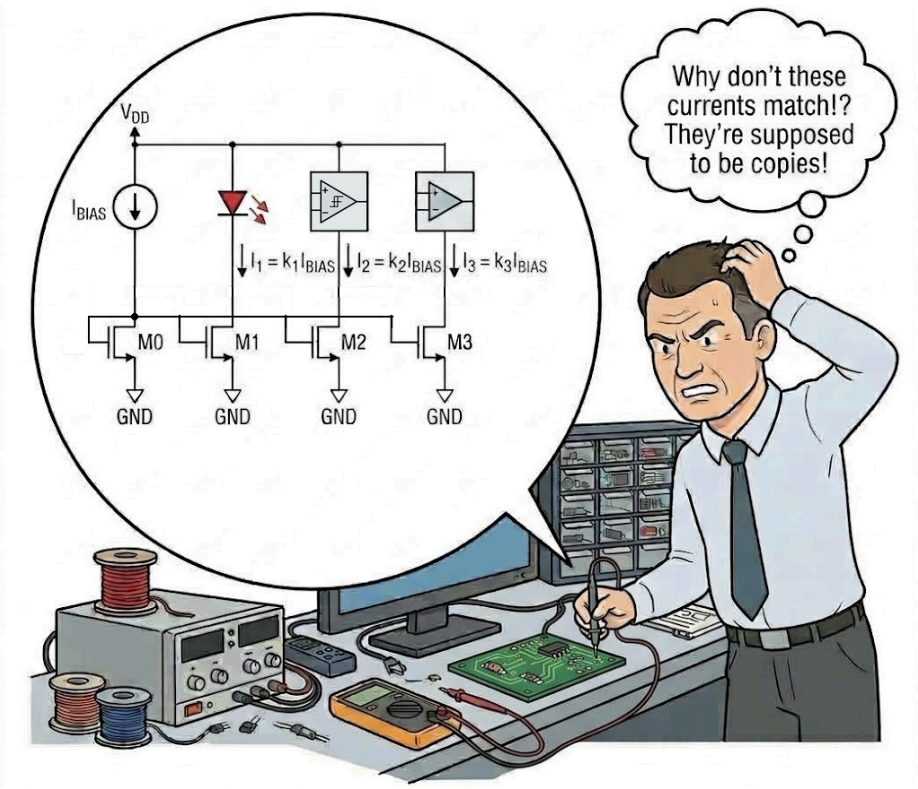
$$G = \frac{R_o - g_m R_o R_F}{(1 + g_m R_o) R_i + R_o + R_F} \qquad f_c = \frac{1}{2\pi} \cdot \frac{(1 + g_m R_o) R_i + R_o + R_F}{R_F C_F [(1 + g_m R_o) R_i + R_o]}$$

Where $Z_F = C_F // R_F$.

Check what the “op point” outputs in the simulation to obtain your values. R_o can be estimated as half of r_o of your input transistor.

Try plugging your values into the equations above and calculate the gain and bandwidth. Are these values closer to your simulated results?

LAB 2: NMOS Current Mirror Matching



Scenario:
Your boss Bob is designing a chip that powers several different circuits at once: an indicator LED, a comparator, and an amplifier stage. Each one needs its own steady bias current, so Bob builds an NMOS current mirror array that copies a single 100 µA reference current into every branch. On paper, each branch should get an exact copy (or an exact multiple) of that reference.
But when Bob measures the currents on his test bench, none of them match what he expected, and each branch is off by a different amount. The transistors are identical copies, the gates are all tied together, and the reference current is extremely steady. Bob is stumped, and he needs you to figure out what's going on.

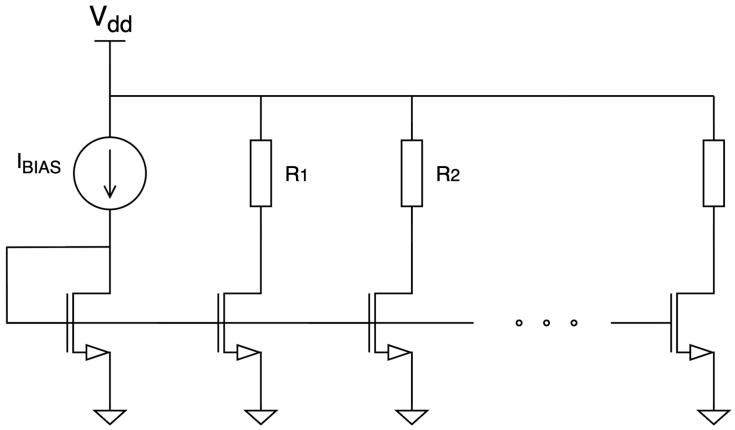
Your Task:
Investigate how differences in drain-to-source voltage affect current matching in an NMOS current mirror array. Calculate resistor values using Ohm's law to establish mirrored currents from a reference current of 100 uA. Then, compare the mirrored output currents to their theoretical values. You must achieve a current mismatch of less than 5%.

Design Constraints:
You must use:

NMOS Current Mirror Array Bias	100 µA

Devices	1x, 2x, 4x, 8x, 16x, 32x (W/L) included in NMOS current mirror array
Resistor	Limited to whatever provided at the event
Supply Voltage	2.5 V

Design Guide:



Start by assuming the current mirrors are ideal. The equation for a current mirror is

$$I_{O,\,nominal} = \frac{(W/L)_2}{(W/L)_1}\, I_{BIAS}$$

where $I_{O,nominal}$ is the nominal current flowing through the drain of the mirroring device of aspect ratio $(W/L)_2$, I_{BIAS} is the current flowing through the drain of the mirrored device of aspect ratio $(W/L)_1$. Using this equation, calculate the nominal current that should flow through the drains of each of the current mirrors. Fill in these results in the “Nominal Current” column of the results table.

Under the ideal assumption, the drain resistance does not affect the drain current (as long as the device stays in saturation).Using the 330 Ω resistor provided, measure the current flowing through each branch by first connecting one side of the resistor to $V_{DD,}$ then probing the multimeter between the other end of the resistor and the appropriate pin on the breadboard. Record each measurement in the “Unmatched Current” column of the results table.

Recall that the ideal current equation for an n-type MOSFET in saturation is

$$I_D = \frac{1}{2}\mu_n C_{ox}(V_{GS} - V_t)^2$$

without taking into account channel length modulation. This effect is accounted for with the additional term, as in

$$I_D = \frac{1}{2}\mu_n C_{ox}(V_{GS} - V_t)^2(1 + \lambda V_{DS})$$

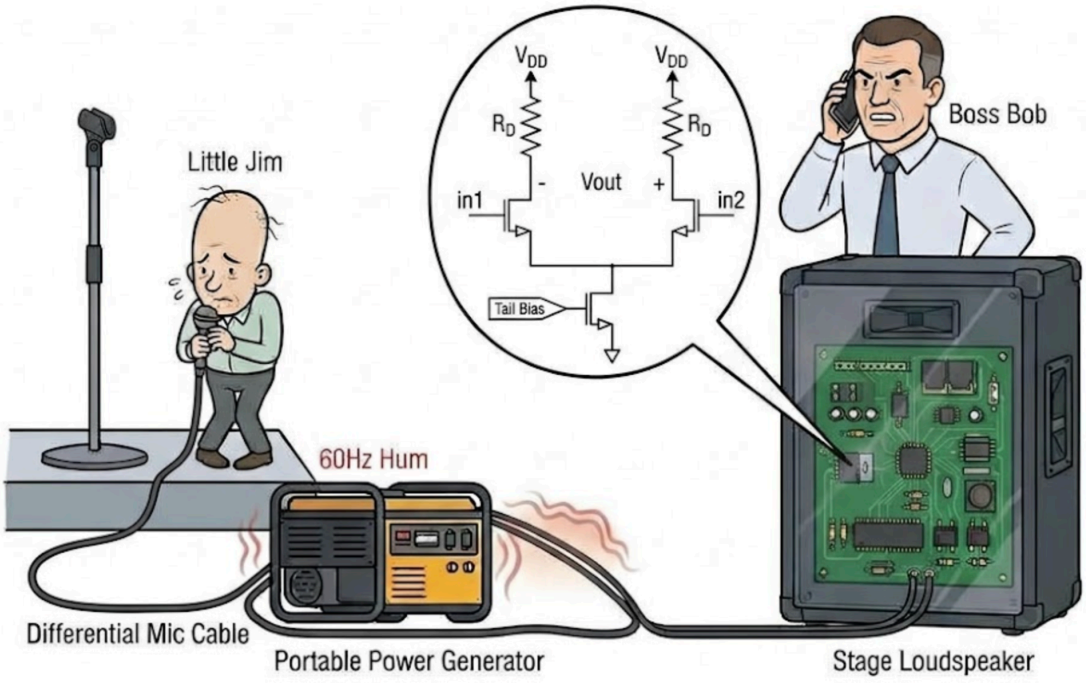
Because channel length modulation causes I_D to also depend on V_{DS}, an equal V_{GS} alone is not enough to guarantee equal current. To minimize current mismatch, the mirroring device should operate with approximately the same V_{DS} as the reference device. Use the simulator to help you choose values of R_D so that the voltage drop across the resistor sets the desired V_{DS} across each of the mirror transistors. After verifying through simulation, record your chosen resistances and corresponding measured current under the “Drain Resistance” and “Matched Current” columns of the results table, respectively. Calculate the errors between the nominal currents and matched currents and record your findings in the results table.

Results Table

Size	Nominal Current [uA]	Unmatched Current [uA]	Drain Resistance [Ω]	Matched Current [uA]	Error [%]
1x					
2x					
4x					
8x					
16x					
32x					

Note:
In the simulator, you may notice that if you perfectly match the V_{DS} of the devices, the mirror copies the exact current (or exact scaled multiple) onto that branch. This won’t happen in the real world. When transistors are fabricated, tiny random variations, such as a few atoms difference in doping or slight differences in the transistor’s dimensions, make no two devices exactly identical. These variations cause a mismatch in threshold voltage and current factor. Foundries have to control their processes with extreme precision to minimize these variations, since these transistors are very small. Designers also play a part, accounting for mismatch when building their circuits and doing the physical layout of the chip.

LAB 3: Differential Amplifier



Scenario:
Your boss, Bob, has booked his band to play the center stage at a shopping mall. The trouble starts at sound check: Little Jim's singing is amazing, but his voice is so quiet that the crowd can barely hear him. Bob calls you in a panic: he needs an amplifier that boosts Jim's voice by 20 dB before the show starts.
There's one more catch. There are no outlets near the stage, so the band is running all of its equipment off a portable power generator, and the cable from Jim's microphone runs right across the generator's power cables, which radiate a strong 60 Hz hum. The hum leaks into both wires of the differential mic cable equally, showing up as common-mode noise. Amplify that, and the loudspeaker will buzz every time Jim pauses. Bob insists on a fully differential amplifier.

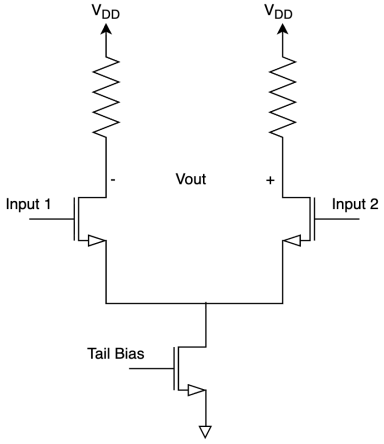
Your Task:
Design a differential amplifier using the MOSbius on-chip NMOS differential pairs that meets the required gain:

Specification Table	
Gain	20 dB

Design Constraints:
You must use the following:

NMOS Current Mirror Array Bias	100 µA
Tail Current Source	2x, 4x, or 8x NMOS in current mirror array
Input Device	1x or 4x W/L Differential Pair NMOS
Drain Resistor	10 kΩ
Supply Voltage	2.5 V
Input Common-Mode Voltage	1.6 V

Design Guideline:



Assuming the transistors have infinite output resistance r_o, the differential gain is

$$A_{v,\,\mathrm{diff}} = g_m R_D$$

Since R_D is fixed at 10 kΩ, the gain specification sets the transconductance g_m your transistors need. Convert the 20 dB spec to V/V, then solve for the required g_m.

Boss Bob was kind enough to include g_m vs. drain current simulation results for both input device sizes. Each current listed is the current through one transistor, so the tail current is twice that value. Choose a device size and a drain current that meet your required g_m.

1x W/L NMOS	**100µA**	**200µA**	**400µA**
gm	740µS	956µS	1.2mS

4x W/L NMOS	**100µA**	**200µA**	**400µA**
gm	1.5 mS	2.193 mS	2.9 mS

Try designing the differential amplifier with the 1x DP NMOS device first (symbol found on the MOSbius software). To reach the required g_m with the 1x device, you would need 400 μA through each transistor. How much voltage does this drop across the 10 kΩ drain resistor? Is there enough voltage left for the transistors to stay in saturation?

Why does the wider 4x DP NMOS device help? Look at the saturation current and transconductance equations below:

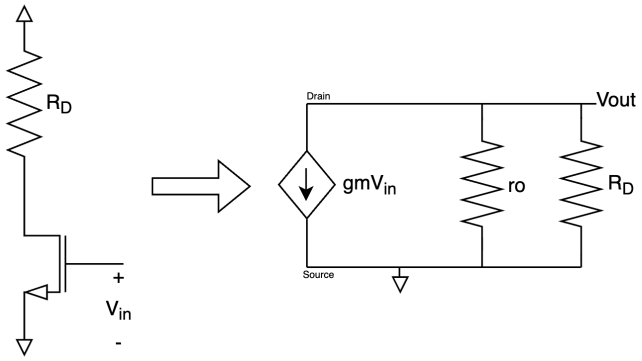
$$I_D = \frac{1}{2} k_n' \frac{W}{L} V_{ov}^2 \qquad\qquad g_m = \frac{2I_D}{V_{ov}}$$

If you increase the width W while keeping I_D the same, the device needs a smaller V_{OV} to carry that current, thus an increase in g_m. In other words, current isn't the only knob designers can tune—a wider device gives more g_m for the same current.

Now design the differential amplifier using the 4x DP NMOS device. Were you able to achieve the required gain? Record your expected gain on the scratch paper.

Simulation Results:
Simulate the differential gain using the math function (V_{outp} - V_{outn}) in the AC analysis of the MOSbius software. Record the gain on the scratch paper. How closely does it match your expected gain?

So far, we have assumed an ideal transistor (infinite output resistance). However, a real transistor has a finite output resistance r_O between its drain and source. To see how this affects the gain, look at the small-signal model below:



The transistor's own output resistance r_O appears in parallel with R_D, so the differential gain becomes:

$$A_{v,\,\mathrm{diff}} = g_m \left(r_o \parallel R_D\right)$$

Check the DC operating point on your schematic and plug its r_O value into the equation above. Does your calculated gain now match the simulation more closely?